

Упражнения

1. $[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A} =$
 $= [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} = \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B} =$$

$$= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

$$[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} + \hat{B}\hat{A}\hat{C} - (\hat{B}\hat{A}\hat{C} + \hat{B}\hat{C}\hat{A}) =$$

$$= \{\hat{A}, \hat{B}\}\hat{C} - \hat{B}\{\hat{A}, \hat{C}\}$$

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{C}, [\hat{A}, \hat{B}]] + [\hat{B}, [\hat{C}, \hat{A}]] = \hat{A}[\hat{B}, \hat{C}] - [\hat{B}, \hat{C}]\hat{A} +$$

$$+ \hat{C}[\hat{A}, \hat{B}] - [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{C}, \hat{A}] - [\hat{C}, \hat{A}]\hat{B} = \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} -$$

$$- \hat{B}\hat{C}\hat{A} + \hat{C}\hat{B}\hat{A} + \hat{C}\hat{A}\hat{B} - \hat{C}\hat{B}\hat{A} - \hat{A}\hat{B}\hat{C} + \hat{B}\hat{A}\hat{C} + \hat{B}\hat{C}\hat{A} - \hat{B}\hat{A}\hat{C} - \hat{C}\hat{A}\hat{B} + \hat{A}\hat{C}\hat{B} = 0$$

2. Пусть $\hat{A}$ - эрмит. оп., пусть $\hat{O} = \hat{A}^\dagger \hat{A}$. Покажем эрмитовость,
 т.е. $\hat{O} = \hat{O}^\dagger$: $\hat{O}^\dagger = (\hat{A}^\dagger \hat{A})^\dagger = \hat{A}^\dagger (\hat{A}^\dagger)^\dagger = \hat{A}^\dagger \hat{A} = \hat{O}$

Пусть эрмит. $|\psi\rangle$: $\langle \psi | \hat{O} | \psi \rangle = \langle \psi | \hat{A}^\dagger \hat{A} | \psi \rangle = \langle \hat{A} \psi | \hat{A} \psi \rangle = \|\hat{A} \psi\|^2 \geq 0$

3. Разл. $e^{\kappa \hat{A}} \hat{B} e^{-\kappa \hat{A}}$ в разг. Теореме при $\kappa=0$

• $\hat{B}$

$$\hat{A} e^{\kappa \hat{A}} \hat{B} e^{-\kappa \hat{A}} - e^{\kappa \hat{A}} \hat{B} \hat{A} e^{-\kappa \hat{A}} = \hat{A}\hat{B} - \hat{B}\hat{A} = [\hat{A}, \hat{B}]$$

$$\hat{A}\hat{A} e^{\kappa \hat{A}} \hat{B} e^{-\kappa \hat{A}} - \hat{A} e^{\kappa \hat{A}} \hat{B} \hat{A} e^{-\kappa \hat{A}} - \hat{A} e^{\kappa \hat{A}} \hat{B} \hat{A} e^{-\kappa \hat{A}} + e^{\kappa \hat{A}} \hat{B} \hat{A} \hat{A} e^{-\kappa \hat{A}} =$$

$$= \hat{A}\hat{A}\hat{B} - \hat{A}\hat{B}\hat{A} - \hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}\hat{A} = \hat{A}[\hat{A}, \hat{B}] - [\hat{A}, \hat{B}]\hat{A} = [\hat{A}, [\hat{A}, \hat{B}]]$$

При $\kappa=1$: $e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + [\hat{A}, [\hat{A}, \hat{B}]] \cdot \frac{1}{2} + \dots$

4. Рассм. $f(\kappa) = e^{\kappa \hat{A}} e^{\kappa \hat{B}}$, $\frac{df}{d\kappa} = \hat{A} e^{\kappa \hat{A}} e^{\kappa \hat{B}} =$
 $= \hat{A} f(\kappa) + e^{\kappa \hat{A}} \hat{B} e^{\kappa \hat{A}} = \hat{A} f(\kappa) + e^{\kappa \hat{A}} \hat{B} e^{-\kappa \hat{A}} f(\kappa) = (\hat{A} + \hat{B} + \kappa \hat{C}) f(\kappa)$
 $0 = [\hat{A}, \hat{C}] = [\hat{B}, \hat{C}]$ по условию

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$$\begin{cases} f(k) = e^{i(\hat{A}+\hat{B}) + \frac{k^2}{2} \hat{C}} \cdot \tilde{C} \\ f(0) = \hat{1} \end{cases} \Rightarrow \tilde{C} = 1$$

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$$f(1) = e^{\hat{A}+\hat{B}+\frac{1}{2}} = e^{\hat{A}} e^{\hat{B}}$$

$$e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}} = e^{\hat{A}+\hat{B}+\frac{1}{2}-\frac{1}{2}} = e^{\hat{A}+\hat{B}}$$

$$e^{\hat{B}} e^{\hat{A}} e^{\frac{1}{2}} = e^{\hat{B}+\hat{A}} \quad \text{uz} \quad [\hat{A}, \hat{B}] = \hat{C} \Rightarrow [\hat{B}, \hat{A}] = -\hat{C}$$

5. Нечто $\hat{U} = e^{i\alpha\hat{A}}$ - унитарное $\Leftrightarrow \hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{1}$

$$\hat{U}^\dagger = e^{(i\alpha\hat{A})^\dagger} = e^{-i\alpha\hat{A}^\dagger} \Leftrightarrow e^{-i\alpha\hat{A}^\dagger} e^{i\alpha\hat{A}} = \hat{1} \Rightarrow$$

$$\Rightarrow \underbrace{e^{-i\alpha\hat{A}^\dagger} e^{i\alpha\hat{A}^\dagger}}_{\hat{1}} e^{i\alpha\hat{A}} = e^{i\alpha\hat{A}^\dagger} \Rightarrow \hat{A} = \hat{A}^\dagger \Rightarrow \hat{A} - \text{эрмитов}$$

6. $\hat{A} F(\hat{B}) \hat{A}^{-1} \stackrel{?}{=} F(\hat{A} \hat{B} \hat{A}^{-1})$

$$F(\hat{B}) = F(0) + \sum \frac{F^{(n)}(0)}{n!} \hat{B}^n$$

$$F(\hat{A} \hat{B} \hat{A}^{-1}) = F(0) + \sum \frac{F^{(n)}(0)}{n!} \hat{A} \hat{B}^n \hat{A}^{-1} = \hat{A} F(\hat{B}) \hat{A}^{-1}$$

7. 1) $\hat{x}|x\rangle = x|x\rangle \Rightarrow \hat{x}^2|x\rangle = \hat{x}(x|x\rangle) = x(\hat{x}|x\rangle) = x^2|x\rangle \Rightarrow$

$$\Rightarrow \hat{x}^n|x\rangle = x^n|x\rangle; \quad \langle x|\hat{x}^n|\tilde{x}\rangle = \tilde{x}^n \langle x|\tilde{x}\rangle =$$

$$= \tilde{x}^n \delta(x-\tilde{x}) = x^n \delta(x-\tilde{x})$$

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2) $\psi_p(x) = \langle x|p\rangle \quad \hat{p}^n|p\rangle = p^n|p\rangle \quad | \langle x|$

$$\langle x|\hat{p}^n|p\rangle = p^n \underbrace{\langle x|p\rangle}_{\psi_p(x)} = (-i\hbar)^n \frac{\partial}{\partial x^n} \psi_p(x) = \int \langle x|\hat{p}^n|\tilde{x}\rangle \underbrace{\langle \tilde{x}|p\rangle}_{\psi_p(\tilde{x})} d\tilde{x}$$

$$\int (-i\hbar)^n \frac{\partial}{\partial x^n} \delta(x-\tilde{x}) \psi_p(\tilde{x}) d\tilde{x} = (-i\hbar)^n \frac{\partial}{\partial x^n} \psi_p(x) \Rightarrow$$

$$\Rightarrow \langle x|\hat{p}^n|\tilde{x}\rangle = (-i\hbar)^n \left(\frac{\partial^n}{\partial x^n} \delta(x-\tilde{x}) \right)$$

$$8. \quad](\hat{x}, F(\hat{p})) = [\hat{x}, \sum_{n=0}^{\infty} a_n \hat{p}^n] = \sum_{n=0}^{\infty} a_n [\hat{x}, \hat{p}^n] \quad \textcircled{=}$$

$$[\hat{x}, \hat{p}^n] = [\hat{x}, \hat{p}^{n-1} \hat{p}] = [\hat{x}, \hat{p}^{n-1}] \hat{p} + \hat{p}^{n-1} [\hat{x}, \hat{p}] = i\hbar(n-1) \hat{p}^{n-1} + i\hbar \hat{p}^{n-1} = i\hbar n \hat{p}^{n-1}$$

$$\textcircled{=} \sum_{n=1}^{\infty} i\hbar n a_n \hat{p}^{n-1} = i\hbar \frac{\partial F(\hat{p})}{\partial \hat{p}}$$

$$2) \quad [\hat{p}, Q(\hat{x})] = -i\hbar \frac{\partial Q(\hat{x})}{\partial \hat{x}} \quad \text{uz} \quad [\hat{p}, \hat{x}^n] = -i\hbar n \hat{x}^{n-1}$$

$$\textcircled{=} \text{m. 3D: } [x_i, x_j] = \delta_{ij} = [\hat{p}_i, \hat{p}_j], \quad [\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij} \hat{1}$$

Задача

$$1. \quad a) \quad \langle x | \hat{O}_I | x' \rangle = \langle x | -x' \rangle = \delta(x+x')$$

$$\langle x | \hat{O}_I^2 | x' \rangle = \int \langle x | \hat{O}_I | \tilde{x} \rangle \langle \tilde{x} | \hat{O}_I | x' \rangle d\tilde{x} = \langle x | -x' \rangle = \delta(x-x')$$

$$b) \quad \langle \psi | \hat{O}_I | \psi \rangle = \int dx dx' \psi^*(x) \delta(x+x') \psi(x') = \int dx \psi^*(x) \psi(-x)$$

$$(\langle \psi | \hat{O}_I | \psi \rangle)^+ = \langle \psi | \hat{O}_I^+ | \psi \rangle = \int dx \psi(x) \psi^*(-x) = \int dx \psi^*(x) \psi(-x) = \langle \psi | \hat{O}_I | \psi \rangle \Rightarrow \hat{O}_I = \hat{O}_I^+ \Rightarrow \text{эрм.}$$

$$\text{Греш. : } \langle \psi | \hat{O}_I^2 | \psi \rangle = \int dx dx' \psi^*(x) \delta(x-x') \psi(x') = \int dx \psi^*(x) \psi(x) = \langle \psi | \psi \rangle \Rightarrow \hat{O}_I^2 = \hat{1}$$

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$$c) \quad \hat{O}_I | \psi \rangle = P | \psi \rangle \quad | \langle x | \Rightarrow \langle x | \hat{O}_I | \psi \rangle = P \psi(x)$$

$$\int dx' \langle x | \hat{O}_I | x' \rangle \langle x' | \psi \rangle = \int dx' \delta(x+x') \psi(x') = \psi(-x) = P \psi(x)$$

$$\hat{O}_I^2 = \hat{1} \Rightarrow P^2 = 1 \Rightarrow P = \pm 1 \text{ (вер. ф.) или } P = -1 \text{ (тот. ф.)}$$

$$d) \quad \langle x | \hat{O}_I | p \rangle = \int dx' \langle x | \hat{O}_I | x' \rangle \langle x' | p \rangle = \int dx' \delta(x+x') e^{i\hbar p x'} = e^{-i\hbar p x} = \langle x | -p \rangle \Rightarrow \hat{O}_I | p \rangle = |-p\rangle$$

$$2. \quad a) \quad \hat{T}(a) = e^{+i\hbar a \hat{p}}, \quad \hat{T}^+(a) = e^{-i\hbar a \hat{p}} = e^{-i\hbar a \hat{p}} = \hat{T}(-a)$$

$$\hat{T}^+(a) \hat{T}(a) = \hat{1}, \text{ т.к. } [-i\hbar a \hat{p}, i\hbar a \hat{p}] = 0 \Rightarrow \hat{T}(a) - \text{унит.}$$

$$b) \hat{T}(a+b) = \hat{T}(a)\hat{T}(b), \text{ т.к. } \left[\frac{i}{\hbar}a\hat{p}, \frac{i}{\hbar}b\hat{p}\right] = 0$$

$$\begin{aligned} c) \langle x | \hat{T}(a) | x' \rangle &= \int dp \langle x | e^{i/\hbar a \hat{p}} | p \rangle \langle p | x' \rangle = \int dp \langle x | \sum_{n=0}^{\infty} \left(\frac{i a \hat{p}}{\hbar}\right)^n \frac{\hat{p}^n}{n!} | x' \rangle = \\ &= \int dp \langle x | e^{i/\hbar a \hat{p}} | p \rangle \langle p | x' \rangle = \int dp e^{i/\hbar a p} \frac{1}{\sqrt{2\pi\hbar}} e^{i p x'/\hbar} e^{-i p x/\hbar} = \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{i/\hbar (x-x'+a)p} dp = \frac{1}{\sqrt{2\pi\hbar}} \cdot \sqrt{2\pi} \delta\left(\frac{x-x'+a}{\hbar}\right) = \delta(x-x'+a) \end{aligned}$$

$$\begin{aligned} d) \langle x | \hat{T}(a) | \psi \rangle &= \int dx' \langle x | \hat{T}(a) | x' \rangle \langle x' | \psi \rangle = \int dx' \delta(x-x'+a) \psi(x') = \psi(x+a) \\ \hat{T}(a)(x) &= \int dx' | x' \rangle \langle x' | \hat{T}(a) | x \rangle = \int dx' | x' \rangle \delta(x'-x+a) = | x-a \rangle \end{aligned}$$

$$e) \hat{T}(a) |\varphi\rangle = p(a) |\varphi\rangle$$

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$$\begin{aligned} \hat{T}(a)\hat{T}(b) |\varphi\rangle &= p(a)p(b) |\varphi\rangle = p(a+b) |\varphi\rangle, \quad p = e^{ika} \\ \int |\hat{T}(a)\varphi(x)|^2 dx &= \int |p(a)\varphi(x)|^2 dx = |p(a)|^2 \underbrace{\int |\varphi(x)|^2 dx}_1 = 1 \Rightarrow \\ \Rightarrow |p(a)|^2 &= 1 \Rightarrow p(a) = e^{ika} \end{aligned}$$

$$\begin{aligned} f) \hat{T}(a) |\varphi\rangle &= e^{ika} |\varphi\rangle \quad | \langle x | \Rightarrow \varphi(x+a) = e^{ika} \varphi(x) \Rightarrow \\ \Rightarrow \text{сво пер. ф. } f: f(x+a) &= f(x) \Rightarrow \varphi_k(x) = e^{ikx} f_k(x) \end{aligned}$$

$$\begin{aligned} \underline{3.} \quad 1) \langle x | \hat{x} | x' \rangle &= x \delta(x-x'), \quad \langle p | \hat{x} | p' \rangle = \int dx dx' \langle p | x \rangle \langle x | \hat{x} | x' \rangle \langle x' | p' \rangle = \\ &= \int dx dx' \langle p | x \rangle x \delta(x-x') \langle x' | p' \rangle = \int dx dx' \frac{1}{\sqrt{2\pi\hbar}} x \delta(x-x') e^{i p' x'/\hbar} e^{-i p x/\hbar} = \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dx \cdot x e^{i x/\hbar} (p'-p) = \frac{1}{\sqrt{2\pi\hbar}} \int dx \frac{\hbar}{i} \frac{\partial}{\partial p'} e^{i x/\hbar} (p'-p) = \\ &= -i\hbar \frac{\partial}{\partial p'} \left(\frac{1}{\sqrt{2\pi\hbar}} \int dx e^{i x/\hbar} (p'-p) \right) = -i\hbar \frac{\partial}{\partial p'} \left(\frac{\sqrt{2\pi}}{\sqrt{2\pi\hbar}} \delta\left(\frac{p'-p}{\hbar}\right) \right) = -i\hbar \frac{\partial}{\partial p'} \delta(p-p') \\ \text{аналогично } \langle p | \hat{x}^2 | p' \rangle &= (-i\hbar)^2 \frac{\partial^2}{\partial p'^2} \delta(p-p') \end{aligned}$$

$$2) \langle x | \hat{p} | x' \rangle = i\hbar \frac{\partial}{\partial x'} \delta(x-x')$$

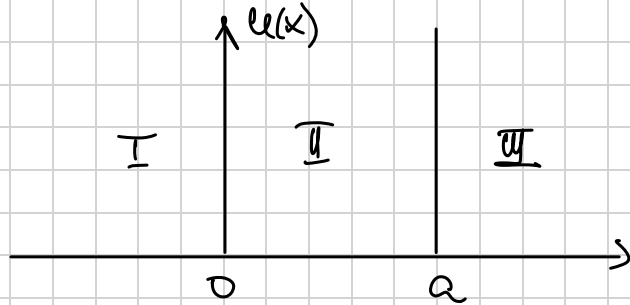
$$\langle p | \hat{p} | p' \rangle = \int dx dx' \frac{1}{\sqrt{2\pi\hbar}} e^{-i p x/\hbar} i\hbar \frac{\partial}{\partial x'} \delta(x-x') e^{i p' x'/\hbar} =$$

$$= -\frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-i p x/\hbar} \hbar \frac{\partial}{\partial x} e^{i p' x/\hbar} = p' \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{i/\hbar (p'-p)x} =$$

$$= p' \cdot \frac{1}{\sqrt{2\pi\hbar}} \cdot \sqrt{2\pi} \delta\left(\frac{p-p'}{\hbar}\right) = p \delta(p-p') ; \quad \langle p | \hat{p}^2 | p' \rangle = p^2 \delta(p-p')$$

$$4. \quad U(x) = \begin{cases} +\infty, & x < 0 \\ 0, & x \in [0, a] \\ -\infty, & x > a \end{cases}$$

$$y.u.: \quad \psi''(x) + \frac{2m}{\hbar^2} (E - U(x)) \psi(x) = 0$$



$$I \text{ и } III: \quad \psi(x) = 0$$

$$II: \quad \psi''(x) + \frac{2mE}{\hbar^2} \psi(x) = 0, \quad k = \frac{\sqrt{2mE}}{\hbar} \Rightarrow \psi(x) = C_1 e^{ikx} + C_2 e^{-ikx}$$

$$\psi(-0) = \psi(+0) \Rightarrow C_1 = -C_2 \Rightarrow \psi(x) = C_1 \sin kx$$

$$\psi(a-0) = \psi(a+0) \Rightarrow C_1 \sin ka = 0 \Rightarrow ka = \pi n \Rightarrow E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2$$

$$\psi_n = \begin{cases} 0, & x \notin [0, a] \\ C \sin \frac{\pi n}{a} x, & x \in [0, a] \end{cases}$$

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$$\text{Нормировка: } \int_{-\infty}^{+\infty} |\psi_n(x)|^2 dx = \frac{a C^2}{\pi n} \left(\frac{\pi n}{2a} - \frac{\sin(\frac{2\pi n}{a} x)}{4} \right) \Big|_0^a = \frac{C^2 a}{2} = 1$$

$$\psi_n = \begin{cases} 0, & x \notin [0, a] \\ \sqrt{\frac{2}{a}} \sin \frac{\pi n}{a} x, & x \in [0, a] \end{cases}$$

$$\text{Измерения: } \langle \Delta x \rangle^2 = \langle \hat{x}^2 \rangle_{\psi_n} - \langle \hat{x} \rangle_{\psi_n}^2$$

$$\begin{aligned} \langle \hat{x} \rangle_{\psi_n} &= \langle E_n | \hat{x} | E_n \rangle = \int dx dx' \psi_n^*(x) x \delta(x-x') \psi_n(x) = \int x |\psi_n(x)|^2 dx = \\ &= \frac{2}{a} \int_0^a x \sin^2 \left(\frac{\pi n}{a} x \right) dx = \frac{a}{2} \end{aligned}$$

$$\langle \hat{x}^2 \rangle_{\psi_n} = \langle E_n | \hat{x}^2 | E_n \rangle = \frac{2}{a} \int_0^a x^2 \sin^2 \left(\frac{\pi n}{a} x \right) dx = a^2 \left(\frac{1}{3} - \frac{1}{2\pi^2 n^2} \right) \Rightarrow$$

$$\Rightarrow \langle \Delta x \rangle^2 = a^2 \left(\frac{1}{12} - \frac{1}{2\pi^2 n^2} \right)$$

$$\begin{aligned} \langle \hat{p} \rangle_{\psi_n} &= \langle E_n | \hat{p} | E_n \rangle = \int dx dx' \psi_n^*(x) i\hbar \frac{\partial}{\partial x'} \delta(x-x') \psi_n(x') = \int dx \psi_n^*(x) (-i\hbar) \frac{\partial}{\partial x} \psi_n(x) \\ &= -\frac{2i\hbar}{a} \int_0^a dx \sin \left(\frac{\pi n}{a} x \right) \cos \left(\frac{\pi n}{a} x \right) \frac{\pi n}{a} dx = \frac{-i\hbar}{2} \sin^2 \left(\frac{\pi n}{a} x \right) \Big|_0^a = 0 \end{aligned}$$

$$\langle \hat{p}^2 \rangle_{\psi_n} = \dots = \frac{2\pi^2 \hbar^2}{a^3} n^2 \int_0^a \sin^2 \left(\frac{\pi n}{a} x \right) dx = \frac{\pi^2 \hbar^2 n^2}{a^2} = \langle \Delta p \rangle^2$$

Coor. репр. $\tilde{r}$: $(\Delta x)^2 (\Delta p)^2 \geq \frac{\hbar^2}{4} > 1$

$$\cancel{\alpha^2} \left(\frac{1}{12} - \frac{1}{2\alpha^2 \hbar^2} \right) \frac{\alpha^2 \hbar^2}{\cancel{\alpha^2}} = \frac{\hbar^2}{4} \left(\frac{\alpha^2 \hbar^2}{3} - 2 \right) \checkmark$$

Рыш: $\frac{\partial \hbar}{\partial x} = \partial k \Rightarrow x = a \frac{k}{n}$

$n=1 \Rightarrow 0$
 $n=2 \Rightarrow \pi/2$
 $n=3 \Rightarrow \pi/3, 2\pi/3$

$n = \frac{a}{\alpha \hbar} \sqrt{2mE} \Rightarrow \rho(E) = \frac{dn}{dE} = \frac{a}{\alpha \hbar} \sqrt{\frac{2m}{E}}$ Рез. $\frac{2a}{\hbar}$

5. а) ψ и ψ' непрерывны

б) I: $\psi(x) = 0$

II: $\psi''(x) - \alpha^2 \psi(x) = 0 \Rightarrow \alpha^2 = \frac{2m|E|}{\hbar^2} \Rightarrow$

$\Rightarrow \psi(x) = C e^{-\alpha x}$

III: $\psi''(x) + (\alpha_0^2 - \alpha^2) \psi(x) = 0, \alpha_0^2 = \frac{2mU_0}{\hbar^2}$

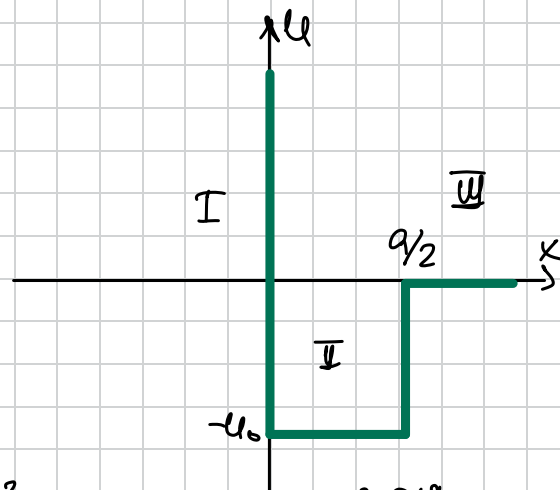
$\psi(x) = A \cos kx + B \sin kx$

из условия ор. $\psi(0) = 0 \Rightarrow E_n = -\frac{\hbar^2 \alpha_n^2}{2m}, C = B \sin k \frac{a}{2} e^{\alpha a/2}$

ген. условие: $\alpha_0 \frac{a}{2} \geq \frac{\pi}{2} \Rightarrow U_0 \geq \frac{\pi^2 \hbar^2}{2ma^2}$

Нормир.: $\langle \psi | \psi \rangle = 1 \Rightarrow \int_0^{a/2} B^2 \sin^2 kx dx + \int_{a/2}^{\infty} C^2 e^{-2\alpha x} dx = 1 \Rightarrow$

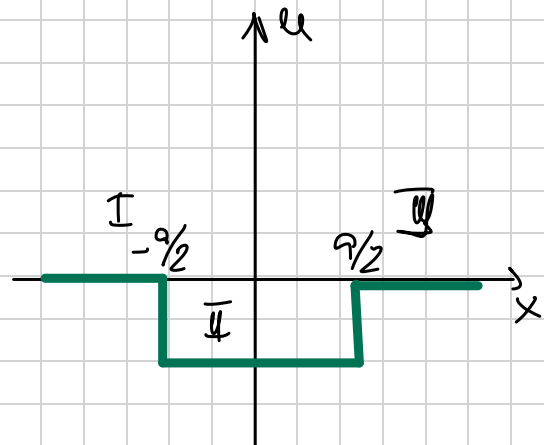
$\Rightarrow B = \left(\frac{a}{4} + \frac{\sin ka}{4k} + \frac{\sin^2 \frac{ka}{2}}{2\alpha} \right)^{-1/2}, C = B \sin k \frac{a}{2} e^{-\alpha a/2}$



6. а) $\psi_I(x) = 1 \cdot e^{ikx} + 2 \cdot e^{-ikx}$
 $\psi_{II}(x) = \beta e^{ikx} + \alpha e^{-ikx}$

II: $\psi'' + \alpha^2 \psi = 0 \Rightarrow \alpha^2 = \frac{2m}{\hbar^2} (E + U_0)$

$\psi_{II}(x) = C_1 e^{i\alpha x} + C_2 e^{-i\alpha x}$



Соединяя: $\begin{cases} \psi_I(-\frac{a}{2}) = \psi_{II}(-\frac{a}{2}) \\ \psi_I'(-\frac{a}{2}) = \psi_{II}'(-\frac{a}{2}) \end{cases} \quad \begin{cases} \psi_{II}(\frac{a}{2}) = \psi_I(\frac{a}{2}) \\ \psi_{II}'(\frac{a}{2}) = \psi_I'(\frac{a}{2}) \end{cases} \Rightarrow$

$$\Rightarrow D = |a|^2 = \frac{4k^2 \omega^2}{4k^2 \omega^2 \cos^2 \omega a + (\omega^2 + k^2)^2 \sin^2 \omega a} = \frac{4k^2 \omega^2}{4k^2 \omega^2 + (\omega^2 - k^2)^2 \sin^2 \omega a}$$

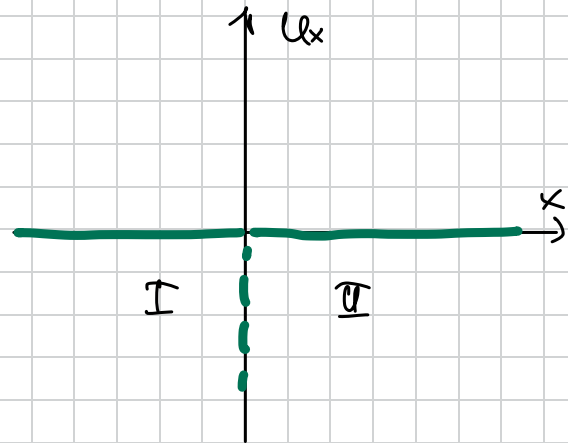
$$R + D = 1 \Rightarrow R = \frac{(\omega^2 - k^2) \sin^2 \omega a}{4k^2 \omega^2 + (\omega^2 - k^2)^2 \sin^2 \omega a}$$

$$S) \eta_{\mu} - \omega_0 \rightarrow +\omega_0 \quad \tilde{\omega}^2 = \frac{2m}{\hbar^2} (\omega_0 - E), \quad \omega = i\tilde{\omega} \Rightarrow \sin \rightarrow \sinh$$

$$7. \quad U(x) = -\frac{\hbar^2}{m} \omega_0 \delta(x)$$

$$\psi''(x) + \frac{2m}{\hbar^2} (E - U(x)) \psi(x) = 0$$

$$\psi''(x) + \left(\frac{2m}{\hbar^2} E + 2\omega_0 \delta(x) \right) \psi(x) = 0$$



$$I: \psi(x) = A e^{\omega x}$$

$$II: \psi(x) = B e^{-\omega x} \quad \omega^2 = \frac{2m|E|}{\hbar^2}$$

$$A = B = C \text{ uz constant}$$

$$\psi'(E) - \psi'(-E) = \int_{-E}^E (\omega^2 - 2\omega_0 \delta(x)) \psi(x) dx = -2\omega_0 \psi(0) + \int_{-E}^E \omega^2 \psi(x) dx$$

$$\psi(+0) - \psi'(-0) = -2\omega_0 \psi(0) \Rightarrow \omega = \omega_0 \Rightarrow E = -\frac{\hbar^2 \omega_0^2}{2m} - \text{Эн. чегз. } \sqrt{\frac{\hbar^2 \omega_0^2}{2m}}$$

$$\text{Норм.: } \int_{-\infty}^{+\infty} |\psi(x)|^2 dx = c^2 \int_{-\infty}^{+\infty} e^{-2\omega_0 |x|} dx = \frac{c^2}{\omega_0} = 1 \Rightarrow c = \sqrt{\omega_0} \Rightarrow \psi(x) = \sqrt{\omega_0} e^{-\omega_0 |x|}$$

$$\langle \hat{x} \rangle_\psi = \langle E | \hat{x} | E \rangle = \int dx dx' \psi^*(x) x \delta(x - x') \psi(x) = \int dx x \psi^*(x) \psi(x) = \int dx x |\psi(x)|^2 dx = \omega_0 \int_{-\infty}^{+\infty} x e^{-2\omega_0 |x|} dx = 0$$

$$\langle \hat{x}^2 \rangle_\psi = \int x^2 |\psi(x)|^2 dx = 2\omega_0 \int_0^\infty x^2 e^{-2\omega_0 x} dx = \frac{1}{2\omega_0^2} = \langle \Delta x \rangle^2$$

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$$\langle \hat{p} \rangle_\psi = \int dx \psi^*(x) (-i\hbar) \frac{\partial}{\partial x} \psi(x) = 0$$

$$\langle \hat{p}^2 \rangle_\psi = \int dx \psi^*(x) (-i\hbar)^2 \frac{\partial^2}{\partial x^2} \psi(x) = -\hbar^2 \int_{-\infty}^{+\infty} dx \omega_0 (\omega_0^2 - 2\omega_0 \delta(x)) e^{-2\omega_0 |x|} = \hbar^2 \omega_0^2$$

$$\text{Coor. Гезз.: } \frac{1}{2\omega_0^2} \cdot \hbar^2 \omega_0^2 = \frac{1}{2} \hbar^2 \geq \frac{1}{4} \hbar^2 - \text{repro.}$$

8. $\hat{H}|E\rangle = E|E\rangle$, $\hat{H} = \frac{\hat{p}^2}{2m} + U(\vec{x})$

$\langle p|\hat{H}|E\rangle = E\langle p|E\rangle = E\psi_E(p)$

$\langle p|\frac{\hat{p}^2}{2m}|E\rangle = \int dp' \langle p|\frac{\hat{p}^2}{2m}|p'\rangle \langle p'|E\rangle = \frac{1}{2m} \int dp' p'^2 \delta(p-p') \psi_E(p') = \frac{p^2}{2m} \psi_E(p)$

$\langle p|U(\vec{x})|E\rangle = \int dx dp' dx' \psi_p^* U(x) \delta(x-x') \psi_E(p') = \frac{1}{2\pi\hbar} \int dx dp' e^{-i/\hbar(p-p')x} U(x) \psi_E(p')$
 $= \frac{1}{2\pi\hbar} \int dp' U(p-p') \psi_E(p')$, т.е. $U(p) = \int dx e^{-i/\hbar px} U(x)$.

Тогда у.у.: $\frac{p^2}{2m} \psi_E(\vec{p}) + \frac{1}{2\pi\hbar} \int d\vec{p}' U(\vec{p}-\vec{p}') \psi_E(\vec{p}') = E \psi_E(\vec{p})$

$U(\vec{p}) = \int d\vec{r} e^{-i/\hbar(\vec{p},\vec{r})} U(\vec{r})$, в сфер. сист. $U(\vec{p}) = U(|\vec{p}|)$

$U(|\vec{p}|) = \int_0^\infty r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi U(r) e^{-i p r \cos\theta / \hbar} = 4\pi \int_0^\infty r^2 U(r) dr \int_{-1}^1 e^{-i p r \cos\theta / \hbar} d\cos\theta$
 $= 4\pi \int_0^\infty r^2 U(r) \frac{\hbar^2}{pr} \cdot \frac{e^{-ipr/\hbar} - e^{ipr/\hbar}}{2i} dr = 4\pi \frac{\hbar}{p} \int_0^\infty r U(r) \sin\left(\frac{pr}{\hbar}\right) dr$

В n-мерном пространстве: $U(p) = \int r^{n-1} U(r) dr \int e^{-i(\vec{p},\vec{r})/\hbar} d\Omega_{n-1}$
 $\frac{2\pi^{n/2}}{\Gamma(\frac{n-1}{2})}$

11. - Селекцион

(12) (13)

(13. $\psi(x,t) = \int dp A(x,p) e^{\frac{i}{\hbar}(Et - px)}$

$A(x,p) = A^{-x^2/4a^2} \delta(p-p_0)$

$\hat{U} + \frac{\partial \psi(x)}{\partial t} \hat{U}$

$\langle p|\hat{A}|p'\rangle$
 ун. переменная
 $|p'\rangle$ - с.б. ор. ун.

В пр. Гейз.: $\frac{\partial \hat{x}^{(H)}}{\partial t} = \left(\frac{\partial \hat{x}^{(H)}}{\partial t}\right) + \frac{i}{\hbar} [\hat{H}, \hat{x}^{(H)}] = \frac{i}{2m\hbar} 2(-i\hbar \hat{p}^{(H)}) = \frac{\hat{p}^{(H)}}{m}$

$\frac{d\hat{p}^{(H)}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{p}^{(H)}] = \frac{i}{2m\hbar} [\hat{p}^{2(H)}, \hat{p}^{(H)}] = 0 \Rightarrow \hat{p}^{(H)} \text{ не мен.} \Rightarrow$

$x^{(H)}(t) = \hat{x} + \frac{\hat{p}}{m} t$; $\psi_0(x, t=0) = \int dp A e^{-x^2/4a^2} \delta(p-p_0) e^{\frac{i}{\hbar} p x} =$
 $= A e^{-x^2/4a^2} e^{\frac{i}{\hbar} p_0 x}$

Нормир.: $\int |\psi(x)|^2 dx = |A|^2 a \sqrt{2\pi} = 1 \Rightarrow A = \frac{1}{\sqrt{a \sqrt{2\pi}}}$

у.у.: $i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$

$\frac{d}{dt} \hat{A}_H(t) = \frac{i}{\hbar} [\hat{H}, \hat{A}_H(t)] + \left(\frac{\partial \hat{A}_S}{\partial t}\right)_H$

$$\langle \hat{x} \rangle = \frac{1}{a\sqrt{2a}} \int dx x e^{-x^2/2a^2} = 0$$

$$\langle \hat{x}^2 \rangle = \frac{1}{a\sqrt{2a}} \int dx x^2 e^{-x^2/2a^2} = a^2$$

$$\langle \hat{x}(t) \rangle = \frac{p_0 t}{m}$$

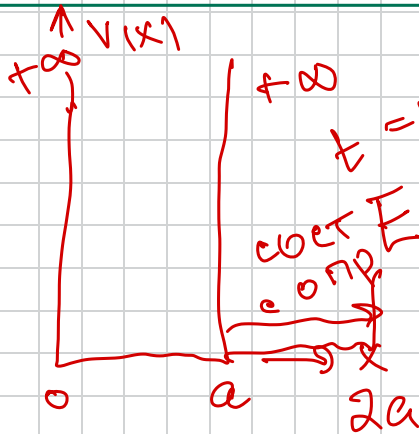
$$\langle \hat{x}^2(t) \rangle = a^2 + \frac{p_0^2 t^2}{m^2} + \frac{\hbar^2 t^2}{8ma^2}$$

$$\langle \hat{p} \rangle = p_0 \quad \langle \hat{p}^2 \rangle = p_0^2 + \frac{\hbar^2}{2a^2} \quad \langle \hat{p}(t) \rangle = p_0 \quad \langle \hat{p}^2(t) \rangle = p_0^2 + \frac{\hbar^2}{2a^2} \quad \geq \frac{\hbar^2}{4}$$

$$\langle \Delta x(t) \rangle^2 = a^2 + \frac{\hbar^2 t^2}{2ma^2}$$

$$\langle \Delta p(t) \rangle^2 = \frac{\hbar^2}{2a^2}$$

$$\langle \Delta x(t) \rangle \langle \Delta p(t) \rangle^2 = \frac{\hbar^2}{2} + \frac{\hbar^4 t^2}{4ma^4}$$



$t=0$ $\psi(x,0)$
 $\cos E_1$
 $\sin E_1$

$t=\Delta t$
 $\psi(x,\Delta t)$
 $\cos E_1$
 $\sin E_1$

$t=T$
 $\psi(x,T)$
 $\cos E_1$
 $\sin E_1$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \quad E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}$$

$$\psi(x,0) = \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) - \text{осн. сост. по расщепл.}$$

$$\text{Для новой шм: } \psi_n(x) = \sqrt{\frac{1}{a}} \sin\left(\frac{\pi n x}{2a}\right) \quad E_n^{2a} = \frac{n^2 \pi^2 \hbar^2}{8ma^2}$$

При нек. возм. в.р. все уст. возм:

$$\psi(x,0+) = \psi_1(x), \quad x \in [0, a]$$

$$\psi(x,0+) = 0, \quad x \in [a, 2a]$$

$$\text{В мом. } t>0: \psi(x,t) = \sum_{n=1}^{\infty} C_n \psi_n(x) e^{-iE_n t/\hbar},$$

$$C_n = \langle \psi_n | \psi(0+) \rangle = \int_0^{2a} \psi_n \psi(x,0+) dx =$$

$$= \int_0^a \frac{1}{\sqrt{a}} \sin\left(\frac{\pi n x}{2a}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) dx + 0 = \frac{-4}{n^2 - 4} \frac{\sqrt{2}}{\pi} \sin\left(\frac{\pi n}{2}\right), \quad n \neq 2$$

$$\text{Norm } n=2 \quad \frac{\sqrt{2}}{a} \int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{\sqrt{2}}{2} \Rightarrow P_2 = |C_2|^2 = \frac{1}{2}$$

$$P_1 = |C_1|^2 = \frac{4\sqrt{2}}{3\pi}$$